Data Mining and Visualization

Final Project Report

Online Shopper Intent Classification

Submitted by

Noam Frei

Uri Lubetzky

Oren Turiel

Course: Data Mining and Visualization

Final Project

Submission Date: June 2026

Contents

1 Introduction

This project focuses on analyzing user behavior in an online shopping platform and building a classification model that predicts whether a user session represents high purchase or engagement intent. Each row in the dataset represents a single browsing session, described by behavioral, contextual, and technical attributes such as the number of pages visited, time spent on different page types, bounce rate, exit rate, traffic source, visitor type, and engagement-related scores.

The project was divided into three main stages. First, we explored the training dataset in order to understand the structure of the data, the distribution of the features, the relationship between attributes, and the behavior of the target variable. Second, we applied preprocessing steps such as cleaning, handling irregular values, transforming features, and preparing the data for machine learning models. Finally, we trained and compared several classification models, tuned their hyperparameters, evaluated them using appropriate metrics, and selected the best-performing model for generating predictions on the provided test dataset.

The target variable in this project is `high_intent`, which indicates whether a session is classified as a high-intent session. Since high-intent sessions are important from a business point of view, the goal of the classification task is not only to achieve high accuracy, but also to correctly identify users who are likely to perform valuable actions on the website.

2 Getting to Know the Data

2.1 Data Loading and Initial Inspection

The first stage of the project was to load the training dataset and inspect its basic structure. The dataset contains user sessions from an online shopping platform, where each row represents one complete session. The goal is to predict the target variable `high_intent`, which indicates whether a session reflects a strong likelihood of meaningful user interaction or conversion.

After loading the file `shopper_train.csv`, we found that the training set contains:

- **9864 rows**, where each row represents one user session.
- **20 columns**, including behavioral features, contextual features, technical features, and the target variable.

The features include page visit counts, time spent on different page categories, engagement indicators, traffic information, visitor type, and session context. This initial inspection confirmed that the dataset contains both numerical and categorical variables, meaning that different preprocessing methods would be required before applying machine learning models.

	num_admin_pages	admin_duration	num_info_pages	info_duration	num_product_pages	product_duration	bounce
0	0	0.0	0	0.0	3	13.000000	0.06
1	0	0.0	0	0.0	1	0.000000	0.20
2	0	0.0	0	0.0	8	101.750000	0.02
3	0	0.0	2	38.2	121	3362.749679	0.02
4	2	2.0	0	0.0	0	NaN	0.10

Figure 1: First rows of the training dataset. Each row represents one user session, and each column describes a behavioral, contextual, or technical attribute of that session.

From the first rows of the dataset, we can already observe several important characteristics. Some sessions contain very little activity, such as sessions with zero administrative or informational pages. Other sessions contain stronger browsing behavior, for example a large number of product pages and long product duration. This shows that the dataset includes sessions with very different levels of user engagement, which is important for the classification task.

2.2 Feature Types and Missing Values

Next, we examined the data types, the number of missing values, the percentage of missing values, and the number of unique values in each column. This step is important because it helps identify which features are numerical, which features are categorical, and which features require cleaning before modeling.

	Data Type	Missing Values	Missing %	Unique Values
num_admin_pages	int64	0	0.00	26
admin_duration	float64	0	0.00	2805
num_info_pages	int64	0	0.00	16
info_duration	float64	0	0.00	1064
num_product_pages	int64	0	0.00	291
product_duration	float64	398	4.03	7480
bounce_rate	float64	1157	11.73	1466
exit_rate	float64	0	0.00	3974
page_value	float64	496	5.03	2082
special_day_score	float64	0	0.00	6
month	object	1460	14.80	10
operating_system	int64	0	0.00	8
browser	int64	0	0.00	12
region	int64	0	0.00	9
traffic_type	int64	0	0.00	20
visitor_type	object	996	10.10	3
is_weekend	int64	0	0.00	2
engagement_score	float64	0	0.00	8909
abandonment_score	float64	0	0.00	4463
high_intent	int64	0	0.00	2

Figure 2: Summary of feature data types, missing values, missing value percentages, and number of unique values.

Most of the numerical behavioral features, such as `num_admin_pages`, `num_info_pages`, `num_product_pages`, `admin_duration`, `info_duration`, `exit_rate`, `engagement_score`, and `abandonment_score`, are stored as integer or floating-point values. Several technical and contextual variables, such as `operating_system`, `browser`, `region`, and `traffic_type`, are represented as integer-coded categorical variables. In contrast, `month` and `visitor_type` are stored as object-type categorical variables.

The missing value analysis showed that most features do not contain missing values. However, several important columns do contain missing data:

- `product_duration`: 398 missing values, approximately 4.03%.
- `bounce_rate`: 1157 missing values, approximately 11.73%.
- `page_value`: 496 missing values, approximately 5.03%.
- `month`: 1460 missing values, approximately 14.80%.

- `visitor_type`: 996 missing values, approximately 10.10%.

These missing values are significant enough that they cannot simply be ignored. Therefore, they must be handled carefully during preprocessing. For numerical features, suitable imputation methods should be considered based on the distribution of each feature. For categorical features such as `month` and `visitor_type`, missing values should be treated in a way that preserves the meaning of the category and avoids introducing artificial numerical relationships.

2.3 Descriptive Statistics of Numerical Features

We then calculated descriptive statistics for the main numerical features. Integer-coded categorical variables such as `operating_system`, `browser`, `region`, and `traffic_type` were excluded from this table, because their numerical values represent category labels rather than continuous quantities.

	count	mean	std	min	25%	50%	75%	max
num_admin_pages	9864.0	2.3160	3.3178	0.0	0.0000	1.0000	4.0000	27.0000
admin_duration	9864.0	80.6495	176.6616	0.0	0.0000	7.0000	94.0000	3398.7500
num_info_pages	9864.0	0.5053	1.2801	0.0	0.0000	0.0000	0.0000	24.0000
info_duration	9864.0	34.8506	143.0302	0.0	0.0000	0.0000	0.0000	2549.3750
num_product_pages	9864.0	31.9672	45.2119	0.0	7.0000	18.0000	38.0000	705.0000
product_duration	9466.0	1210.7741	1995.7962	0.0	181.0000	601.7792	1475.9182	63973.5222
bounce_rate	8707.0	0.0227	0.0493	0.0	0.0000	0.0032	0.0171	0.2000
exit_rate	9864.0	0.0434	0.0490	0.0	0.0143	0.0253	0.0500	0.2000
page_value	9368.0	5.9088	18.3766	0.0	0.0000	0.0000	0.0000	361.7637
special_day_score	9864.0	0.0612	0.1985	0.0	0.0000	0.0000	0.0000	1.0000
engagement_score	9864.0	15.2683	18.6523	0.0	5.0755	9.7516	18.1499	291.0692
abandonment_score	9864.0	0.0329	0.0479	0.0	0.0080	0.0158	0.0329	0.2000

Figure 3: Descriptive statistics of the main numerical features in the training dataset. Integer-coded categorical variables were excluded from this summary.

The descriptive statistics reveal several important patterns. First, many page-count and duration-based features are highly skewed. For example, the median number of product pages is 18, while the maximum value is 705. Similarly, the median product duration is approximately 602 seconds, while the maximum value is above 63,000 seconds. This indicates that most users spend a moderate amount of time browsing, while a small number of sessions contain extremely large values.

A similar pattern appears in the duration features. For example, `admin_duration` and `info_duration` both have a median close to zero, but their maximum values are much higher. This suggests that many users do not visit administrative or informational pages at all, while a smaller group of users spends a significant amount of time on these pages.

The feature `page_value` is also highly skewed. Its median is zero, while its maximum value is above 361. This is meaningful because many sessions have no direct commercial value, while a smaller number of sessions are associated with high-value pages. Since `page_value` is likely related to conversion behavior, it may be an important predictor for `high_intent`.

The rate-based features, such as `bounce_rate`, `exit_rate`, and `abandonment_score`, are bounded between 0 and 0.2. These features describe disengagement behavior, so they are expected to be useful for separating low-intent sessions from high-intent sessions. In contrast, `engagement_score` has a much wider range, from 0 to approximately 291, showing that users differ greatly in their level of interaction with the website.

Overall, the initial statistical analysis suggests that preprocessing will need to handle missing values, skewed numerical distributions, categorical variables, and features with very different scales. “‘

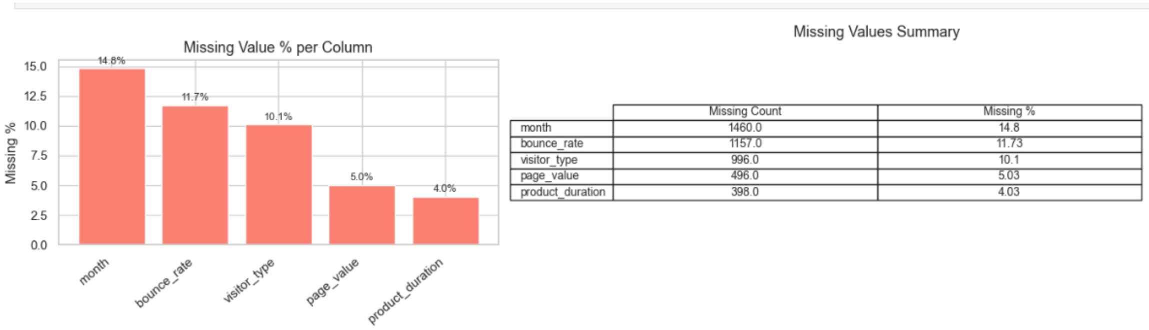
“‘latex

2.4 Data Quality Assessment

Before choosing preprocessing methods, we performed a data quality assessment. The goal of this step was to identify missing values, irregular feature distributions, strong skewness, and possible outliers. This is important because classification models are sensitive to the way the data is represented, and poor handling of missing or extreme values may lead to unreliable results.

2.4.1 Missing Values

We first examined missing values at the column level. The analysis showed that missing values appear only in a small subset of the features, while most columns are complete.



	Missing Count	Missing %
month	1460.0	14.8
bounce_rate	1157.0	11.73
visitor_type	996.0	10.1
page_value	496.0	5.03
product_duration	398.0	4.03

Figure 4: Missing value percentage and missing value count for each column containing missing data.

The feature with the highest missing percentage is `month`, with 1460 missing values, which is approximately 14.8% of the dataset. This is followed by `bounce_rate`, with 1157 missing values, or approximately 11.7%. The feature `visitor_type` contains 996 missing values, approximately 10.1%. The numerical features `page_value` and `product_duration` contain smaller missing percentages of approximately 5.0% and 4.0%, respectively.

This means that the missing values are not negligible, especially for `month`, `bounce_rate`, and `visitor_type`. Therefore, deleting all rows with missing values would remove a large amount of useful data. Instead, the missing values should be handled during preprocessing using appropriate imputation methods according to the feature type.

We also examined missing values at the row level. Most rows do not contain missing values, but a significant number of rows contain one or more missing entries.

Figure 5: Number of missing values per row in the training dataset.

The row-level analysis showed that 6056 rows have no missing values, while 3153 rows contain exactly one missing value. In addition, 611 rows contain two missing values, and 44 rows contain three missing values. Since most incomplete rows contain only one missing value, it is reasonable to preserve these rows and fill the missing entries instead of removing them.

2.4.2 Skewness of Numerical Features

Next, we examined the skewness of the numerical features. Skewness measures how asymmetric a distribution is. A value close to zero indicates a relatively balanced distribution, while a large positive value indicates that most observations are concentrated near small values, with a long tail toward large values.

	Feature	Skewness
0	product_duration	7.755150
1	info_duration	7.599689
2	page_value	6.313180
3	admin_duration	5.743064
4	num_product_pages	4.266238
5	num_info_pages	4.159568
6	engagement_score	4.084522
7	special_day_score	3.311363
8	bounce_rate	2.892041
9	abandonment_score	2.632243
10	exit_rate	2.127351
11	num_admin_pages	1.921935

```
Features with |skewness| > 1 are candidates for log transformation.
```

Figure 6: Skewness values of the numerical features. Features with high positive skewness are candidates for logarithmic transformation.

The skewness analysis shows that many numerical features are strongly right-skewed. The most skewed features are `product_duration`, `info_duration`, `page_value`, and `admin_duration`. These features describe time spent on pages or commercial value, so the skewness is expected: many users spend little or no time on certain page types, while a small number of users spend a much longer time.

The high skewness suggests that applying a logarithmic transformation may be useful for some features. A log transformation reduces the effect of extremely large values while preserving the order of the data. This can help models that are sensitive to scale and extreme values, such as KNN and logistic regression.

2.4.3 Distributions of Numerical Features

To better understand the numerical features, we plotted their raw distributions.

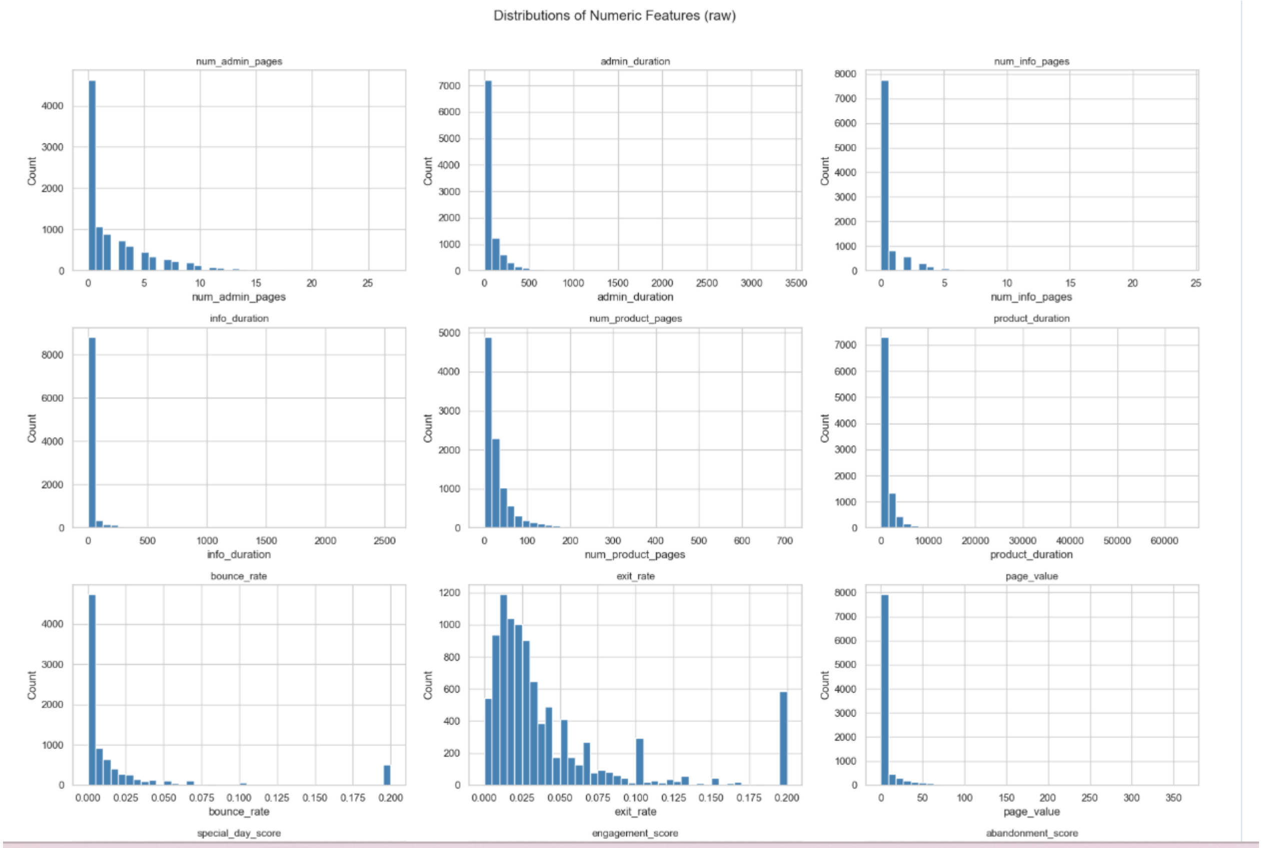


Figure 7: Raw distributions of the main numerical features. Many features are concentrated near zero and have long right tails.

The histograms confirm the skewness results. Several features, such as `admin_duration`, `info_duration`, `product_duration`, and `page_value`, have most of their values close to zero, with a small number of very large observations. This indicates that many sessions contain little activity in these categories, while a smaller number of sessions show much stronger browsing behavior.

The feature `num_product_pages` is also right-skewed. Most sessions contain a relatively small or moderate number of product page visits, but some sessions contain hundreds of product page views. This may represent users with high browsing activity, repeated product comparison, or unusual session behavior.

The rate-based variables, such as `bounce_rate`, `exit_rate`, and `abandonment_score`, are bounded and have different behavior from the duration features. These variables are important because they describe user disengagement. For example, high bounce or exit rates may indicate lower intent, while lower values may suggest that the user continued browsing.

2.4.4 Outlier Analysis

We also used boxplots to inspect potential outliers in the numerical features.

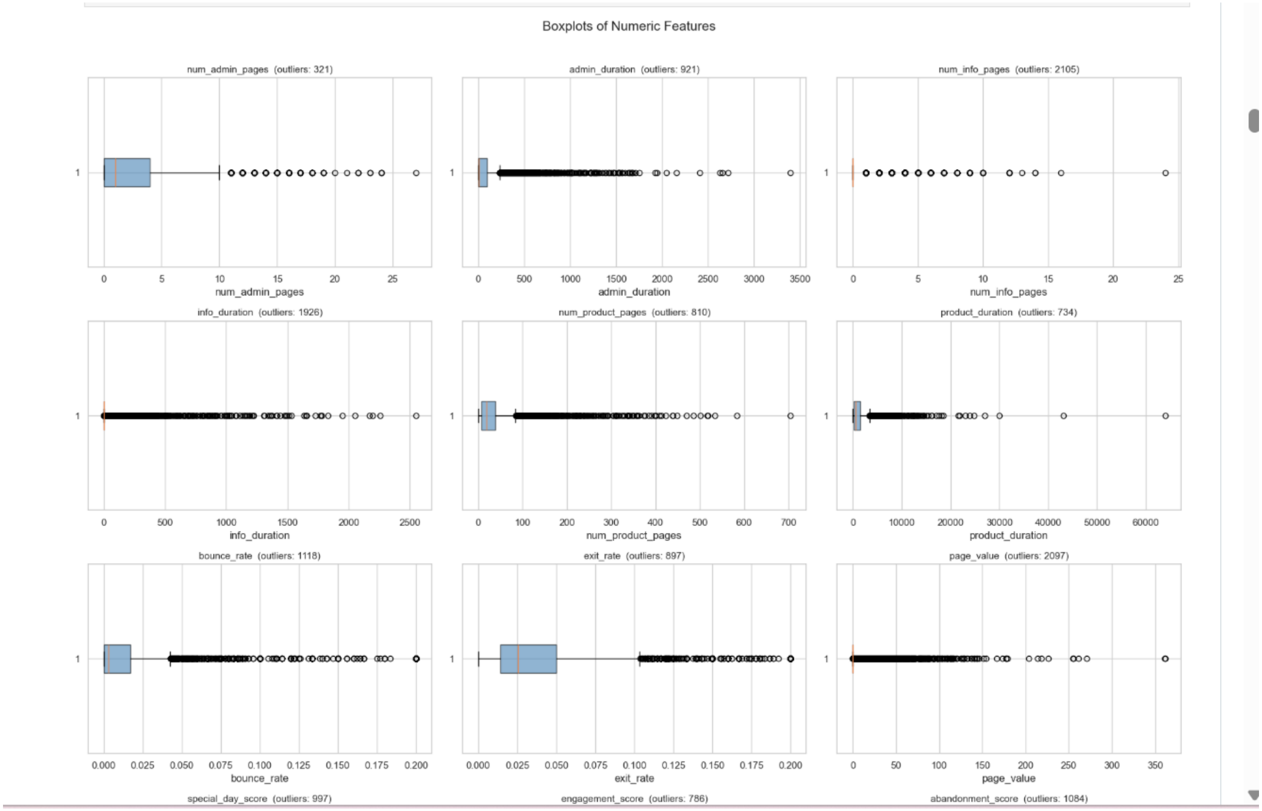


Figure 8: Boxplots of numerical features. The plots show many upper outliers, especially in duration, page count, and value-related features.

The boxplots show that many numerical features contain upper outliers. This is especially clear in `admin_duration`, `info_duration`, `product_duration`, `page_value`, and `num_product_pages`. However, these values should not automatically be treated as errors. In this dataset, large values may represent real user behavior, such as a user spending a long time browsing products or visiting many product pages during one session.

Therefore, instead of removing these observations immediately, we decided to handle them carefully during preprocessing. In particular, logarithmic transformations and scaling are more appropriate than simple deletion, because they reduce the influence of extreme values while preserving useful information.

2.4.5 Categorical Feature Distributions

We then examined the distributions of the categorical and integer-coded categorical features.

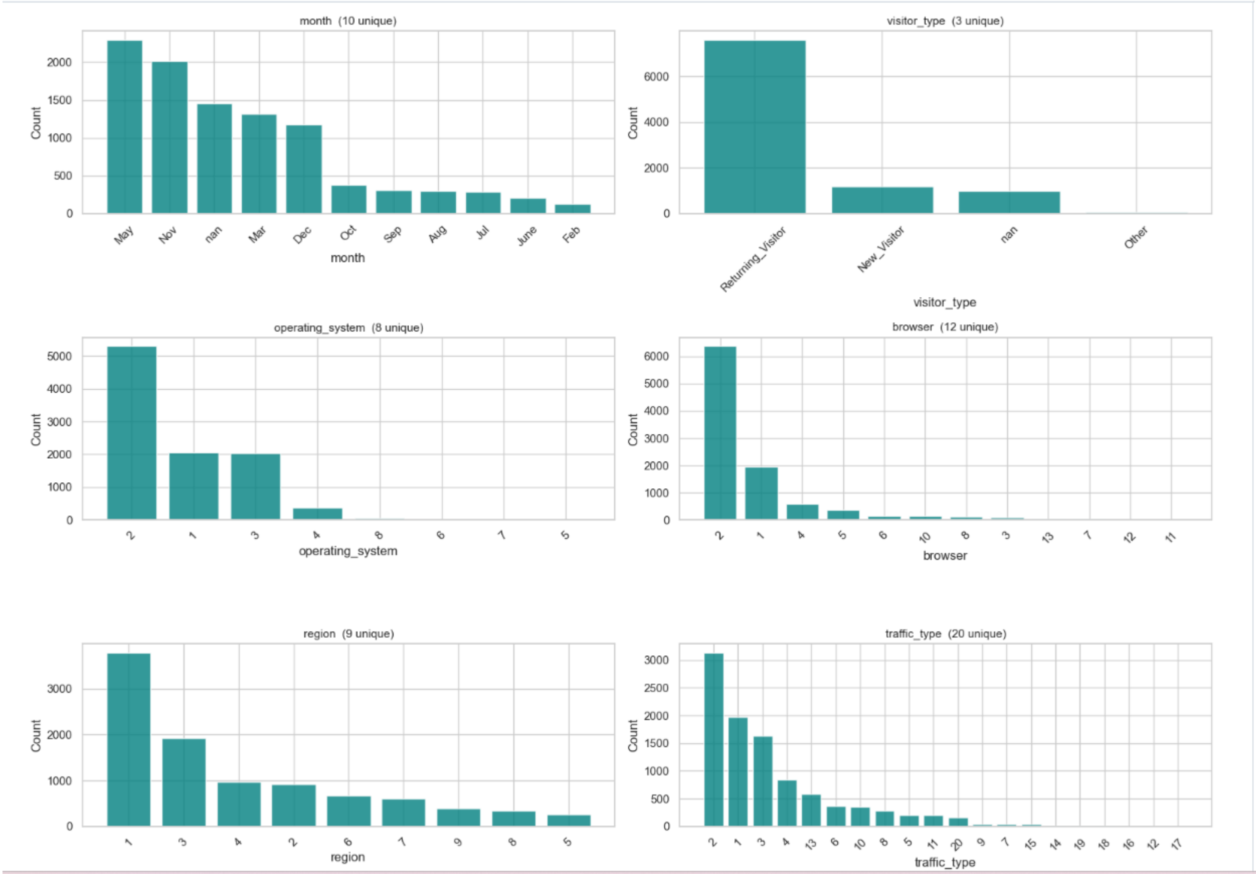


Figure 9: Distributions of categorical and integer-coded categorical features.

The categorical feature distributions are not uniform. For example, most sessions belong to a small number of months, with May and November appearing most frequently. The feature `visitor_type` is dominated by returning visitors, while new visitors form a much smaller group. Similarly, `operating_system`, `browser`, `region`, and `traffic_type` are concentrated around a few common categories.

This observation is important for preprocessing. Since these features represent categories rather than continuous numerical quantities, they should not be treated as regular numerical variables. For example, browser category "2" is not necessarily larger or better than browser category "1". Therefore, categorical encoding is required before applying most machine learning models.

2.4.6 Target Variable Distribution

Finally, we examined the distribution of the target variable, `high_intent`.

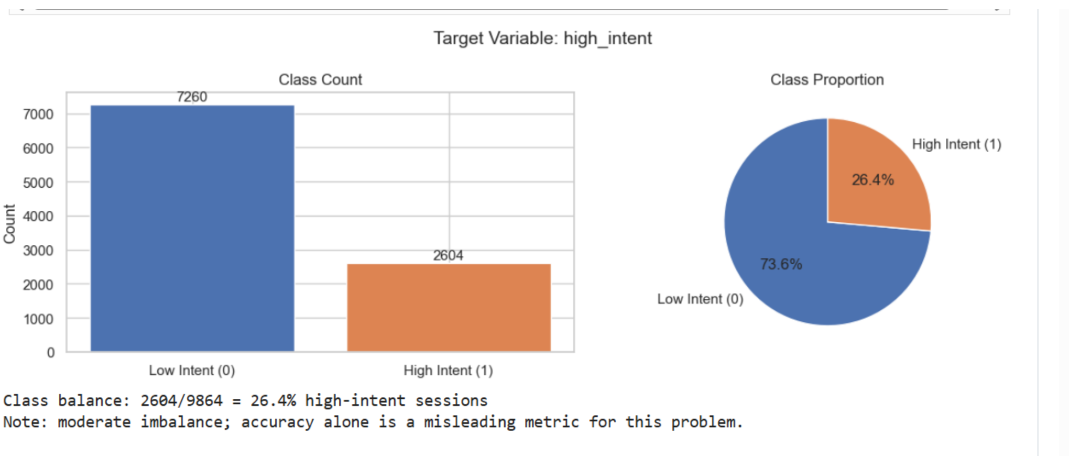


Figure 10: Distribution of the target variable `high_intent`.

The target distribution shows that the dataset is moderately imbalanced. Out of 9864 sessions, 7260 are labeled as low intent, while 2604 are labeled as high intent. This means that high-intent sessions represent approximately 26.4% of the data, while low-intent sessions represent approximately 73.6%.

This class imbalance is important for model evaluation. If we only use accuracy, a model may achieve a seemingly good score by mainly predicting the majority class. However, the business goal is to identify high-intent users, so evaluation metrics such as precision, recall, F1-score, and ROC-AUC are also needed. This observation directly affects the classification stage, where we compare models using metrics that better reflect performance on both classes.

3 Preprocessing

4 Classification

5 Final Summary